

# AV2 Codec ISO Media File Format Binding

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## This version:

<https://aomediacodec.github.io/av2-isobmff/v1.0.0.html>

## Issue Tracking:

[GitHub](#)

## Editors:

TBD

TBD

## Warning

This specification is still at draft stage and should not be referenced other than as a working draft.

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## Abstract

This document specifies the storage format for [AV2] bitstreams in [ISOBMFF] tracks as well as in [CMAF] files.

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## § 1. Terminology

The key words "**MUST**", "**MUST NOT**", "**REQUIRED**", "**SHALL**", "**SHALL NOT**", "**SHOULD**", "**SHOULD NOT**", "**RECOMMENDED**", "**NOT RECOMMENDED**", "**MAY**", and "**OPTIONAL**" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

All of the text of this specification is normative except sections explicitly marked as non-normative, examples, and notes.

Examples in this specification are introduced with the words "for example" or are set apart from the normative text with `class="example"`, like this:

### EXAMPLE 1

This is an example of an informative example.

Informative notes begin with the word "Note" and are set apart from the normative text with `class="note"`, like this:

**NOTE:** This is an example of an informative note.

## § 2. Bitstream features overview

An [AV2 bitstream](#) is composed of a sequence of [OBUs](#) (Open Bitstream Units), grouped into [Temporal Units](#). A Temporal Unit consists of a [Temporal Delimiter OBU](#) and all following OBUs up to, but not including, the next Temporal Delimiter OBU. Each Temporal Unit is associated with a single presentation time.

Every OBU begins with a 1- or 2-byte header that carries the [obu\\_type](#) and the temporal layer identifier [obu\\_tlayer\\_id](#). When the OBU header extension is present (i.e. when [obu\\_header\\_extension\\_flag](#) is equal to 1 in that header), two additional layer identifiers are carried: the embedded layer identifier [obu\\_mlayer\\_id](#) and the extended layer identifier [obu\\_xlayer\\_id](#). Depending on its type, an OBU can carry configuration information (e.g., a [Sequence Header OBU](#) or [Layer Configuration Record OBU](#)), metadata, or coded video data.

A Temporal Unit **MAY** contain multiple coded pictures but only one is output. AV2 defines a number of coded-frame OBU types, including [Closed Loop Key OBUs](#), [Open Loop Key OBUs](#), regular and leading tile-group OBUs, [Switch OBUs](#), [OBU\\_RAS\\_FRAME](#), [Bridge Frame OBUs](#), and TIP (Temporally Interpolated Prediction) frames.

Frames that can be decoded without reference to other frames are [Key Frames](#) (carried in a [Closed Loop Key OBU](#) or an [Open Loop Key OBU](#)) and [Intra frames](#). All other coded frames - including [Inter Frames](#), [Switch Frames](#), RAS frames, Bridge frames, TIP frames, and frames with [show\\_existing\\_frame](#) equal to 1 - have coding dependencies on other frames in the same or previous Temporal Units.

A [Closed Random Access](#) is the random access process that applies to an extended layer whose first coded frame is a [Key Frame](#) carried in a [Closed Loop Key OBU](#). It starts a new coded video sequence for the extended layer; the layer's reference frame buffers are invalidated and every subsequent frame of that layer can be decoded. This is the primary random-access mechanism.

An [Open Random Access](#) is the random access process that applies to an extended layer whose first coded frame is a [Key Frame](#) carried in an [Open Loop Key OBU](#). During sequential decoding, an [Open Random Access](#) point does not start a new coded video sequence; however, when a decoder initiates decoding at that point, the decoding process is treated as if it were the start of a new coded video sequence for the extended layer. [Leading frames](#) that follow an [Open Random Access](#) point **MAY** reference frames that precede it, and are discarded when decoding begins at the [Open Random Access](#) point.

In addition to Key Frames, AV2 provides two further mechanisms that support random access or stream switching: a RAS frame (OBU\_RAS\_FRAME) is an inter-predicted frame that uses long-term reference frames and enables random access or switching without inserting a full key frame; a [Switch Frame](#) (OBU\_SWITCH) is an inter-predicted frame that enables switching between representations at a given point, typically in adaptive streaming scenarios.

AV2 supports three scalability dimensions: temporal layers (identified by [obu\\_tlayer\\_id](#)), embedded layers (identified by [obu\\_mlayer\\_id](#)), and extended layers (identified by [obu\\_xlayer\\_id](#)). The reserved [obu\\_xlayer\\_id](#) value GLOBAL\_XLAYER\_ID (equal to 31, as defined in [\[AV2\]](#)) denotes global scope and is used by OBUs that do not belong to any specific extended layer (such as the [Temporal Delimiter OBU](#)). A bitstream that uses a single distinct non-global value of [obu\\_xlayer\\_id](#) is a [Singlestream](#); a bitstream that uses two or more distinct non-global values is a [Multistream](#).

AV2 defines two distinct [Metadata OBU](#) types for carrying metadata: OBU\_METADATA\_SHORT, using the [metadata short OBU syntax](#), and OBU\_METADATA\_GROUP, using the [metadata group OBU syntax](#). A separate OBU type carries film grain synthesis parameters; its presence is indicated by the [film\\_grain\\_params\\_present](#) flag in the [Sequence Header OBU](#).

The [Sequence Header OBU](#) also carries the bitstream's profile, level, and - when applicable - tier, as defined in Annex A of [\[AV2\]](#).

The [Sequence Header OBU](#) carries a [monotonic\\_output\\_order\\_flag](#) that defines the output mode of the associated coded video sequence.

When [monotonic\\_output\\_order\\_flag](#) is equal to 1, the output order of coded output frame equals their decoding order within the coded video sequence. Within a [sample](#), the composition time (CTS) of the associated output frame is equal to the [sample](#)'s decoding time (DTS); the [sample](#) **MAY** additionally contain hidden frames.

When [monotonic\\_output\\_order\\_flag](#) is equal to 0, the output order of coded output frames **MAY** differ from their decoding order. The composition time of each [sample](#) is then derived from a ['ctts](#) box that signals the offset between the [sample](#)'s decoding time (DTS) and the composition time of its associated output frame. As in the monotonic case, a [sample](#) **MAY** additionally contain hidden frames.

**NOTE:** Non-monotonic output mode is new in AV2; AV1 supports only the monotonic mode.

## § 3. Basic Encapsulation Scheme

This section describes the basic data structures used to signal encapsulation of [AV2 bitstreams](#) in [\[ISOBMFF\]](#) containers.

There are two locations where **OBUs** can be found in a file conformant to this specification:

- Some **High-Level Syntax OBUs**, such as the **Sequence Header OBU**, are carried in the **AV2CodecConfigurationBox** as described in § 3.2 **AV2 Sample Entry and Configuration**. This corresponds to "availability through external means" as described in the [AV2] specification. These **OBUs** apply to all **samples** of the corresponding **AV2SampleEntry**.
- The other **OBUs** are stored in **samples**. This corresponds to "presence in the bitstream" as described in the [AV2] specification.

An **AV2 sample** is the union of a **sample** and the **OBUs** carried in the **AV2CodecConfigurationBox** of the corresponding **AV2SampleEntry**, forming exactly one **Temporal Unit** of the **AV2 bitstream**.

**NOTE:** This specification refers to **samples** as defined in [ISOBMFF], which are not to be mistaken with **samples** or **sample values** as defined in the [AV2] specification.

## § 3.1. General Requirements & Brands

A file conformant to this specification satisfies the following:

- It SHALL conform to the normative requirements of [ISOBMFF]
- It SHALL have the **'av02'** brand among the compatible brands array of the **FileTypeBox**
- It SHALL contain at least one track using an **AV2SampleEntry**, possibly transformed by encryption as specified in § 5 **Common Encryption**
- It SHOULD indicate a structural ISOBMFF brand among the compatible brands array of the **FileTypeBox**, such as **'iso6'**
- For a given **AV2 sample**, the concatenation of the **OBUs** carried in the **AV2CodecConfigurationBox** of the corresponding **AV2SampleEntry** first, followed by the **OBUs** stored in the **sample**, SHALL form a conformant [AV2] **Temporal Unit**

## § 3.2. AV2 Sample Entry and Configuration

### § 3.2.1. AV2 Sample Entry

The **AV2SampleEntry** extends **VisualSampleEntry** and signals a track carrying **AV2 samples**.

```
class AV2SampleEntry extends VisualSampleEntry('av02') {  
    AV2CodecConfigurationBox config;  
}
```

An **AV2SampleEntry** SHALL contain exactly one **AV2CodecConfigurationBox**. An **AV2SampleEntry** SHALL contain a **'colr'** box with the **'nclx'** colour type. Each value among **colour primaries**, **transfer characteristics**, **matrix coefficients**, and **full\_range\_flag** carried in that **'colr'** box:

- SHALL equal the corresponding effective value of the AV2 bitstream carried by the track if the latter is anything but 2 (Unspecified);
- SHALL take precedence over the corresponding effective value of the AV2 bitstream carried by the track.

An [AV2SampleEntry](#) SHALL contain exactly one 'pixi' box with the least significant bit of 'px\_flags' enabled.

**NOTE:** An [AV2SampleEntry](#) may contain additional sub-boxes permitted by [\[ISOBMFF\]](#) for a [VisualSampleEntry](#), such as 'pasp', 'clli', 'mdcv' or 'btrt'.

### § 3.2.2. AV2 Codec Configuration Box

The **AV2CodecConfigurationBox** carries the decoder initialisation data required to decode the [AV2 samples](#) of a track.

```
aligned(8) class AV2CodecConfigurationBox extends Box('av2C')
{
    unsigned int(8) reserved1 = 0;
    unsigned int(8) config_obus_count_minus1;
    for (i = 0; i < config_obus_count_minus1 + 1; ++i) {
        unsigned int(8) config_obu[];
    }
}
```

reserved SHALL be set to 0. Readers SHALL ignore its value.

config\_obus\_count\_minus1 is the number of config\_obu minus one.

config\_obu is a byte array carrying one OBU and its length as described in [§ 3.2.3 config\\_obu](#).

### § 3.2.3. config\_obu

Each config\_obu SHALL carry a single OBU preceded by a `leb128()` `num_bytes_in_obu` value giving the byte length of that OBU, as specified in the [Length delimited bitstream format](#) from Annex B of [\[AV2\]](#).

config\_obu SHALL NOT contain OBUs of the types forbidden there as stated in [Figure 3.1](#).

| <b>AV2 OBU type</b>                                                                                                                           | <b>Be present in<br/>AV2CodecConfigurationBox<br/>(in config_obu)</b> | <b>Be present in a <a href="#">sample</a></b>                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| OBU_SEQUENCE_HEADER                                                                                                                           | <b>SHALL</b>                                                          | <b>SHALL NOT</b>                                                                              |
| OBU_TEMPORAL_DELIMITER                                                                                                                        | <b>SHALL NOT</b>                                                      | <b>SHALL NOT</b> (the <a href="#">AV2 sample</a> boundary already delimits the temporal unit) |
| OBU_MULTI_FRAME_HEADER                                                                                                                        | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_CLOSED_LOOP_KEY                                                                                                                           | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_OPEN_LOOP_KEY                                                                                                                             | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_LEADING_TILE_GROUP                                                                                                                        | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_REGULAR_TILE_GROUP                                                                                                                        | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_METADATA_SHORT or OBU_METADATA_GROUP,<br>when its contents do not change for the<br>duration of the active <a href="#">AV2SampleEntry</a> | <b>MAY</b>                                                            | <b>SHOULD NOT</b>                                                                             |
| OBU_METADATA_SHORT or OBU_METADATA_GROUP,<br>otherwise                                                                                        | <b>SHOULD NOT</b>                                                     | <b>MAY</b>                                                                                    |
| OBU_SWITCH                                                                                                                                    | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_LEADING_SEF                                                                                                                               | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_REGULAR_SEF                                                                                                                               | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_LEADING_TIP                                                                                                                               | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_REGULAR_TIP                                                                                                                               | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_BUFFER_REMOVAL_TIMING                                                                                                                     | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_LAYER_CONFIGURATION_RECORD                                                                                                                | <b>MAY</b>                                                            | <b>SHALL NOT</b>                                                                              |
| OBU_ATLAS_SEGMENT                                                                                                                             | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_OPERATING_POINT_SET                                                                                                                       | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_BRIDGE_FRAME                                                                                                                              | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_MSDO                                                                                                                                      | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_RAS_FRAME                                                                                                                                 | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
| OBU_QUANTIZATION_MATRIX                                                                                                                       | <b>SHALL NOT</b>                                                      | <b>MAY</b>                                                                                    |
|                                                                                                                                               | <b>MAY</b>                                                            | <b>SHOULD NOT</b>                                                                             |

|                                                                                                               |                   |                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBU_FILM_GRAIN, when its contents do not change for the duration of the active <a href="#">AV2SampleEntry</a> |                   |                                                                                                                                                                               |
| OBU_FILM_GRAIN, otherwise                                                                                     | <b>SHOULD NOT</b> | <b>MAY</b>                                                                                                                                                                    |
| OBU_CONTENT_INTERPRETATION                                                                                    | <b>MAY</b>        | <b>SHALL NOT</b>                                                                                                                                                              |
| OBU_PADDING                                                                                                   | <b>SHALL NOT</b>  | <b>SHALL NOT</b> (byte-level padding within a track is achieved using ISOBMFF mechanisms, e.g. <a href="#">‘free’/‘skip’</a> boxes or <a href="#">sample</a> size signalling) |

**Figure 3.1: List of AV2 OBU types and where they can be found**

The first config\_obu **SHALL** carry a [Sequence Header OBU](#). For a [Multistream](#), config\_obu **SHALL** contain the [Sequence Header OBU](#) of each extended layer together with the stream’s global configuration OBUs, in the order specified by the [ordering of OBUs in a temporal unit](#) in [AV2]; its first OBU may therefore be a [Layer Configuration Record OBU](#) rather than a [Sequence Header OBU](#).

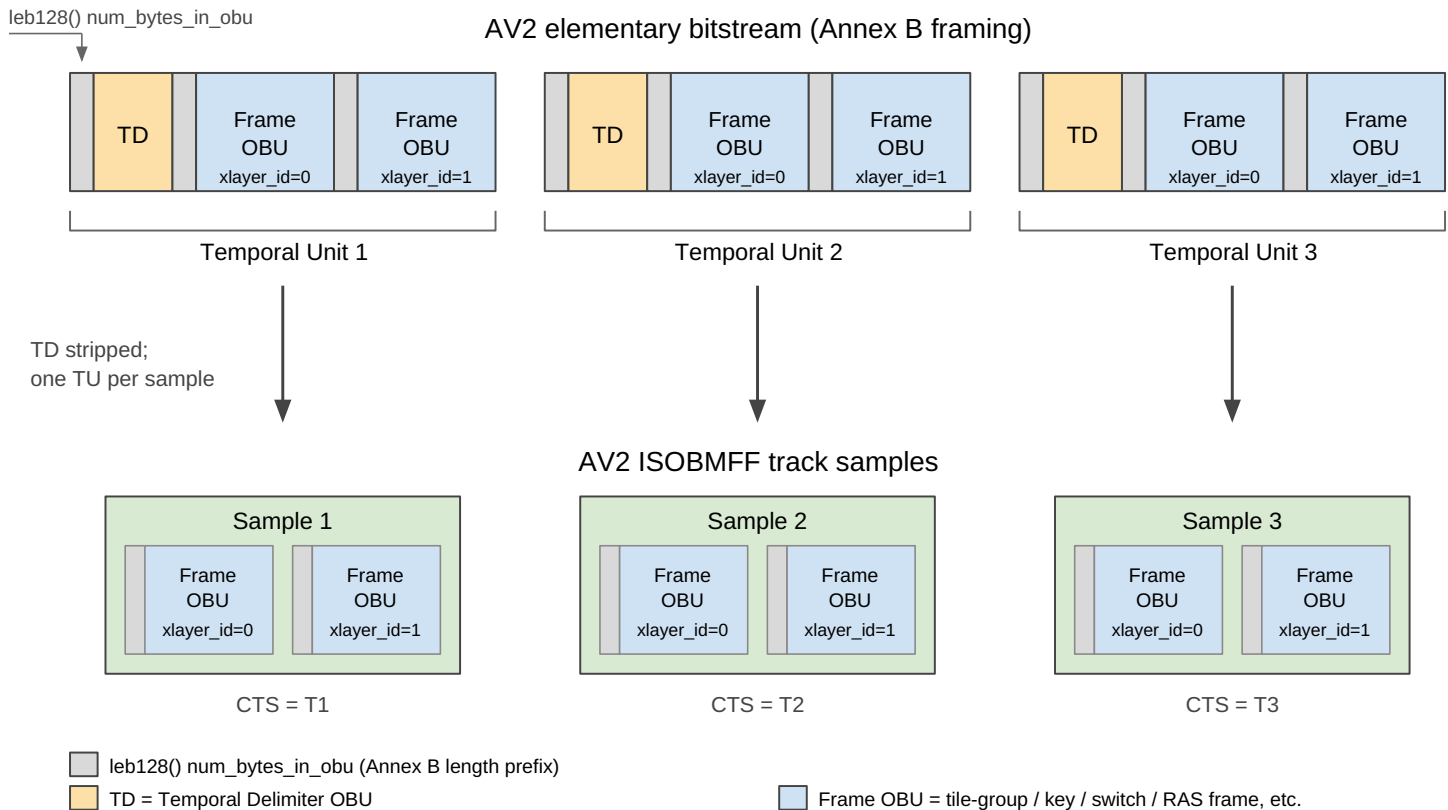
**ISSUE 1** Editor’s note: for a [Multistream](#), config\_obu contains a [Sequence Header OBU](#) for each extended layer. The impact on multi-layer use cases — including how a reader determines the active decoder configuration per extended layer from multiple Sequence Header OBUs — needs investigation. Also how profile / level is signalled to decoder API for layers. etc.

## § 3.3. AV2 Sample Format

### § 3.3.1. General

The [AV2 sample](#) boundary delimits the temporal unit.

Each OBU within a [sample](#) SHALL be framed using the [Length delimited bitstream format](#) specified in Annex B of [AV2]: each OBU is preceded by a leb128() num\_bytes\_in\_obu field giving the byte length of the OBU that follows. [Figure 3.2](#) illustrates this mapping.



**Figure 3.2: Example mapping of AV2 temporal units into the ISOBMFF sample format.**

### § 3.3.2. OBUs in samples

The **OBUs** within a **sample** SHALL appear in the order specified by the **ordering of OBUs in a temporal unit** in [AV2], with the following constraint:

- the **OBUs** within a **sample** SHALL NOT contain **OBUs** of the types forbidden there as stated in Figure 3.1.

**ISSUE 2** Editor's note: the Content Interpretation OBU (OBU\_CONTENT\_INTERPRETATION) carries picture-level interpretation metadata (for example, colour description) that overlaps ISOBMFF container-level signalling such as the ColourInformationBox. Whether to carry it in the track or replace it entirely with container-level signalling is to be resolved in this version.

The handling of the **Temporal Delimiter OBU** depends on how a track's **samples** are consumed. When the **samples** are passed to an AV2 decoder API that already treats each **sample** as one **Temporal Unit**, a **Temporal Delimiter OBU** is not required. When the **samples** are reconstructed into an **AV2 bitstream** in the **Length delimited bitstream format** specified in Annex B of [AV2], or into any other carriage that does not itself delimit temporal units, a **Temporal Delimiter OBU** (framed with its `leb128()` length prefix) SHALL be prepended ahead of each **sample** that does not already start with one.

### § 3.3.3. Sync sample

An **AV2 sample** is a **sync AV2 sample** if, for every **coded extended layer unit** present in the **AV2 sample**, the first coded frame OBU of that coded extended layer unit has `obu_type` equal to `OBU_CLOSED_LOOP_KEY`. Such

an [AV2 sample](#) is a [Closed Random Access](#) point for the carried bitstream and corresponds to a SAP of type 1 or 2 as defined in Annex I of [\[ISOBMFF\]](#).

A [sync AV2 sample](#) SHALL be self-sufficient for starting decoding at that point for all layers: every OBU referenced by the coded frame OBUs of the [sync AV2 sample](#) SHALL be available either by being carried in the [AV2CodecConfigurationBox](#) of the active [AV2SampleEntry](#) or by being present in the [sample](#), as required by the [availability of HLS OBUs](#) in [\[AV2\]](#).

**NOTE:** In ISOBMFF terms, sync samples are signalled using the SyncSampleBox ([‘stss’](#)) in unfragmented tracks, or by setting `sample_is_non_sync_sample` equal to 0 in fragmented tracks. When every sample in a track is a sync sample, the SyncSampleBox is omitted in unfragmented tracks (per [\[ISOBMFF\]](#)), and in fragmented tracks this is expressed through the default sample flags of the MovieExtendsBox ([‘trex’](#)) or TrackFragmentHeaderBox ([‘tfhd’](#)) with no per-sample override.

## § 4. CMAF AV2 track format

TBD

## § 5. Common Encryption

TBD

## § 6. Codecs Parameter String

TBD

## § Index

## § Terms defined by this specification

[av02](#), in § 3.1

[AV2 sample](#), in § 3

[sync AV2 sample](#), in § 3.3.3

[AV2CodecConfigurationBox](#), in § 3.2.2

[AV2SampleEntry](#), in § 3.2.1

## § Terms defined by reference

[AV2] defines the following terms:

AV2 bitstream  
availability of HLS OBUs  
Bridge Frame OBU  
Closed Loop Key OBU  
Closed Random Access  
coded extended layer unit  
High-Level Syntax  
Inter Frame  
Intra frame  
Key Frame  
Layer Configuration Record OBU  
Leading frame  
Length delimited bitstream format  
metadata group OBU syntax  
Metadata OBU  
metadata short OBU syntax  
Multistream  
OBU  
obu\_header\_extension\_flag  
obu\_mlayer\_id  
obu\_tlayer\_id  
obu\_type  
obu\_xlayer\_id  
Open Loop Key OBU  
Open Random Access  
ordering of OBUs in a temporal unit  
sample  
sample value  
Sequence Header OBU  
show\_existing\_frame  
Singlestream  
Switch Frame  
Switch OBU  
Temporal Delimiter OBU

Temporal Unit

[ISOBMFF] defines the following terms:

btrt  
clli  
colr  
ctts  
iso6  
mdcv  
nclx  
pasp  
pixi  
sample  
VisualSampleEntry

## § References

### § Normative References

#### **[AV2]**

*AV2 Bitstream & Decoding Process Specification*. Standard. URL: <https://av2.aomedia.org/v1.0.0/index.html>

#### **[CMAF]**

*Information technology — Multimedia application format (MPEG-A) — Part 19: Common media application format (CMAF) for segmented media*. Standard. URL: <https://www.iso.org/standard/71975.html>

#### **[ISOBMFF]**

*Information technology — Coding of audio-visual objects — Part 12: ISO base media file format*. Standard. URL: <https://www.iso.org/standard/83102.html>

#### **[RFC2119]**

*Key words for use in RFCs to Indicate Requirement Levels*. Standard. URL: <https://www.rfc-editor.org/info/rfc2119>

#### **[RFC8174]**

*Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words*. Standard. URL: <https://www.rfc-editor.org/info/rfc8174>